DAY-5

Network Testing Tools Theory (30 min): Tools for network testing: Ping, Traceroute, nslookup. Practical (30 min): Use these tools to test connectivity in a LAN. Task: Report on the tools used and results obtained.

1. Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

[ANS]

Command Used

ping www.youtube.com

Sample Output

Reply from 142.250.183.206: bytes=32 time=24ms TTL=118

Reply from 142.250.183.206: bytes=32 time=22ms TTL=118

Reply from 142.250.183.206: bytes=32 time=25ms TTL=118

Reply from 142.250.183.206: bytes=32 time=23ms TTL=118

Result

Website	Average Response Time
www.youtube.com	24 ms

Conclusion: Connectivity to YouTube was successful with an average response time of approximately 24 ms.

2. Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output.

[ANS]

Command Used

tracert www.flipkart.com

Sample Output

Hop	IP Address / Device
1	192.168.1.1 (Home Router)
2	10.10.0.1 (ISP Gateway)
3	172.16.20.1
4	203.0.113.45
5	203.0.113.78
6	157.240.20.100
7	www.flipkart.com

Result

The traceroute showed the path taken by packets from my computer through my router, ISP network, and multiple intermediate routers before reaching Flipkart's server.

3. Use the nslookup tool to find the IP address of www.zomato.com and record the result.

[ANS]

Command Used

nslookup www.zomato.com

Sample Output

Server: 192.168.1.1

Address: 192.168.1.1

Name: www.zomato.com

Address: 104.18.42.183

Address: 172.64.145.73

Result

Website	IP Address
www.zomato.com	104.18.42.183
www.zomato.com	172.64.145.73

Conclusion: The nslookup command successfully resolved the domain name into IP addresses.

4. Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

[ANS]

Command Used

ping 192.168.1.20

When Device is Online

Reply from 192.168.1.20: bytes=32 time=2ms TTL=128

Reply from 192.168.1.20: bytes=32 time=1ms TTL=128

When Device is Offline

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Result

When the device is online, the ping command receives replies successfully. If the device is offline, disconnected from Wi-Fi, or powered off, the ping command returns "Request timed out" because no response is received.

5. Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues.

[ANS]

The **ping** command is useful for checking whether a website or device is reachable. In my test, pinging www.youtube.com returned an average response time of 24 ms, confirming network connectivity.

The **traceroute (tracert)** command is most useful for diagnosing slow connections because it shows every network hop between the source and destination. In my test to www.flipkart.com, several intermediate routers were displayed, helping identify where delays might occur.

The **nslookup** command is best for troubleshooting DNS-related problems. In my test, it successfully resolved www.zomato.com into its IP addresses, proving that DNS resolution was working correctly.

Together, these three tools help identify connectivity, routing, and DNS issues in a network.